CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

Minimal sets of generators

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Abstract

A minimal set of generators of a right module MR over an associative ring R with identity is a subset X of M that generates M, but for any $x \in X$, $X \setminus \{x\}$ generates a proper submodule of MR. Unlike in the case of R a division ring, minimal sets of generators have little to do with free sets of generators of M or maximal free subsets of M. Most of the talk will deal with the case of R a local ring. For R a local ring, minimal sets of generators are related to projective covers of the module MR. Hence right perfect rings appear in a natural way in the setting. The results presented are due to Ercolanoni and myself ("Projective covers over local rings", to appear in Ann. Mat. Pura Appl.), Chwe and Neggers, Herden, Hrbek and Ruzicka, Nashier and Nichols.